Data Mining Report II

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1 Introduction

The report thus far has focused on the exploration and classification of prostate cancer data, based on a dataset of 6033 gene expressions. The first part included detailed descriptive analysis, data

visualizations, and the application of feature selection methods (Wilcoxon test and ANOVA) in conjunction with classification using various machine learning models. It was demonstrated that despite subtle differences in the expression of individual genes, multivariate approaches allowed for effective differentiation between cancerous and non-cancerous samples.

In the second part of this report, using the same dataset, we will delve deeper into unsupervised machine learning techniques. We will focus on cluster analysis with quality assessment, which will enable the discovery of natural structures within the data without prior labels. Additionally, we will investigate the application of selected dimensionality reduction methods in connection with both classification and cluster analysis, aiming to identify the most informative gene combinations and their impact on improving data detectability and interpretability.

2 Clusterization

In this chapter, cluster analysis is presented as an unsupervised machine learning technique used to group a set of objects in such a way that objects in the same group (called a cluster) are more similar to each other than to those in other groups. Its primary purpose is to discover natural groupings or underlying patterns within data without prior knowledge of labels or categories.

2.1 Clustering methods

For the analysis, we will employ three distinct clustering algorithms (with variations):

1. **K-Means:** It is one of the most popular clustering algorithms, characterized by its simplicity and computational efficiency, especially for large datasets. It operates iteratively, aiming to partition n observations into k predefined clusters. The algorithm starts by randomly selecting k points as initial centroids (cluster centers). In each iteration, every data point is assigned to the nearest centroid, and then the centroids are recomputed as the mean of all points belonging to that cluster. This process repeats until cluster assignments no longer change or a maximum number of iterations is reached.

2. **AGNES** is an example of a hierarchical agglomerative clustering algorithm, meaning it builds a hierarchy of clusters "from the bottom up." Initially, each data point is treated as a separate, single-element cluster. The algorithm then iteratively merges the two closest clusters, forming a new, larger cluster. This process continues until all points are in one large cluster or a desired hierarchy level is achieved. The result of AGNES is a dendrogram, which visually represents the cluster merging process. In this analysis, we will utilize different linkage methods to define the distance between clusters:

   - **Average linkage:** Connects clusters based on the average distance between all pairs of points from each cluster.
   - **Complete linkage:** Connects clusters based on the greatest distance between any pair of points from each cluster.
   - **Single linkage:** Connects clusters based on the smallest distance between any pair of points from each cluster.

3. **Gaussian Mixture:** A Gaussian Mixture Model (GMM) is a probabilistic clustering approach that models the data as a combination of multiple Gaussian distributions. Each cluster

corresponds to a Gaussian component defined by its mean and covariance. We consider two types of covariance structures in this analysis:

- **Diagonal covariance:** Each Gaussian has an axis-aligned ellipsoidal shape, allowing different variances in each dimension but assuming no correlation between features.
- **Spherical covariance:** Each Gaussian is constrained to be a hypersphere, assuming identical variance in all directions and no feature correlation, resulting in isotropic clusters.

Furthermore, we will assess the quality of the obtained clustering methods.

2.2 Internal Indices

Internal cluster validity indices are methods for evaluating the quality of a clustering result based solely on the inherent properties of the data and the clusters themselves, without using external information about true labels. They measure compactness (how closely related the points are within a cluster) and separation (how well clusters are separated from each other).

2.2.1 Silhouette Score

The Silhouette Score measures how well each object fits into its own cluster and how poorly it fits into neighboring clusters. For each data point, a Silhouette Score value is calculated, ranging from -1 to 1. A value close to 1 indicates that the point is well-matched to its own cluster and far from other clusters. A value close to 0 means that the point is near the decision boundary between two clusters, while a value close to -1 suggests that the point may have been incorrectly assigned to its cluster. The average Silhouette Score across all points provides an overall assessment of the clustering quality.

2.2.2 Davies-Bouldin Index

The Davies-Bouldin Index is a cluster validity measure that relies on the idea that well-clustered clusters should be compact and well-separated from each other. For each cluster, the ratio of the within-cluster variance (a measure of dispersion of points within the cluster) to the distance between cluster centroids is calculated. The final index value is the average of these ratios across all clusters. A lower Davies-Bouldin index value indicates better clustering, meaning more compact clusters that are better separated from each other. It is a dimensionless measure, which facilitates the comparison of results from different clustering algorithms.

2.3 External Indices

External cluster validity indices are measures used to evaluate the quality of clustering by comparing the obtained clusters with a previously known "ground truth" structure of the data (class labels). Unlike internal indices, they require external reference information about the correct grouping.

2.3.1 Adjusted Rand Index (ARI)

The Adjusted Rand Index (ARI) is a measure of similarity between two partitions of a set of data. In the context of cluster evaluation, ARI compares the assignment of objects to clusters made by the algorithm with their true class labels. The ARI value ranges from -1 to $+1$. A value of $+1$ indicates a perfect match between the clustering and the true labels (i.e., the clustering perfectly reflects the true classes). A value of 0 indicates that the agreement is random, as if assignments were purely by chance. Negative values (rare in practice) suggest worse agreement than random. ARI is corrected for chance, meaning that random clustering has an expected value close to 0, which makes it more reliable for comparing results from different algorithms and for varying numbers of clusters.

2.3.2 Purity

Purity is a simple and intuitive measure for evaluating clustering quality that also requires true class labels for the data. For each cluster formed by the algorithm, Purity determines the predominant class within that cluster. It then sums the number of points that were correctly assigned to the predominant class in each cluster and divides it by the total number of points in the dataset. The Purity value ranges from 0 to 1, where 1 indicates a perfect clustering in which each cluster contains only points of a single true class. Despite its simplicity and ease of interpretation, Purity has a drawback: it is heavily dependent on the number of clusters. The more clusters there are, the higher the tendency to achieve a high Purity value, even if the clustering is not meaningful (e.g., each cluster contains only one data point).

3 Results for Clustering

Clustering was performed using the bootstrap method. For each clustering algorithm, 50 bootstrap iterations were carried out. Within each iteration, the algorithm was applied with the number of clusters varying from 2 to 14. For each cluster count, the previously mentioned evaluation metrics were computed. After completing 50 bootstrap iterations, the mean value of each metric was calculated for every cluster number.

3.0.1 Results for Internal Indices

Figure 1 illustrates the comparison of Silhouette and Davies-Bouldin Index scores for K-Means, AGNES algorithms with different linkage methods (average linkage, complete linkage, single linkage) and Gaussian Mixture with different linkage methods (diagonal, spherical) as a function of the number of clusters k, ranging from 2 to 14.

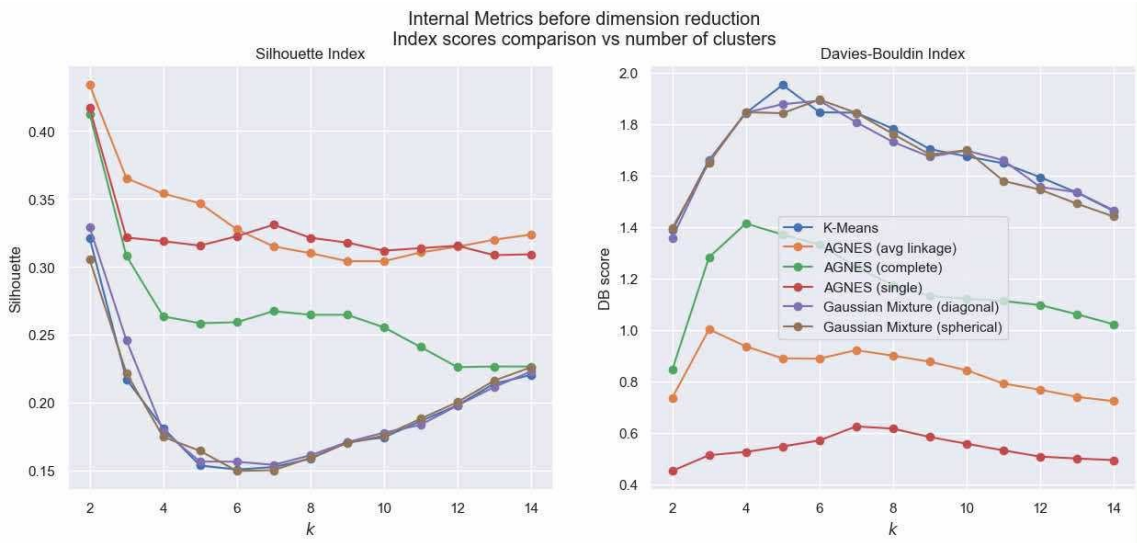

Figure 1: Index scores comparison for a given number of clusters K

1. **Silhouette Index**:

   - **AGNES (avg linkage)** achieves the highest Silhouette Index score at $K = 2$, obtaining a value above 0.40. This indicates the most distinct and compact cluster structure for this method at this number of clusters. As K increases, the score steadily declines, suggesting that adding more clusters deteriorates the natural grouping in the data.
   - **AGNES (single linkage)** also achieves a high Silhouette Index value at $K = 2$ (approximately 0.30), however its score drops much faster for $K > 3$. This linkage method is known for its tendency to create elongated, chain-like clusters, which might explain the performance drop with increasing complexity.
   - **AGNES (complete linkage)** obtains its best Silhouette Index score at $K = 3$ (approximately 0.25), although these values remain fairly stable within the range of $0.24 - 0.26$ for higher K values. This suggests moderately cohesive, though less compact, clusters compared to average linkage.
   - **K-Means** achieves a Silhouette Index score of approximately 0.30 at $K = 2$, with a noticeable drop-off for $K > 3$. The overall decreasing trend indicates K-Means' limited ability to form well-separated clusters in this dataset, likely due to the high-dimensional and non-spherical structure of gene expression data, which violates K-Means' assumptions.
   - **Gaussian Mixture (diagonal)** Initially scores around 0.33 for K=2, and then sharply declines, remaining at a low level (below 0.20) for most K values. This suggests that the Gaussian Mixture model with a diagonal covariance matrix is not effective in identifying well-separated clusters in this dataset.
   - **Gaussian Mixture (spherical)** Behaves very similarly to Gaussian Mixture (diagonal), with an initial score of about 0.33 for K=2 and a rapid decline thereafter. This indicates

that the assumption of spherical clusters also does not fit the inherent data structure, leading to low Silhouette Index scores for a larger number of clusters.

2. **Davies-Bouldin Index**:

   - **AGNES (single linkage)** initially increases, reaching a maximum value of about 1.85 for K=5 or K=6, and then decreases. Low Davies-Bouldin Index values are desirable, so higher values in this case indicate less optimal clustering, which contradicts the observations for the Silhouette Index. This might be due to the Davies-Bouldin Index being more sensitive to cluster density and size than the Silhouette Index.
   - **AGNES (single linkage)** exhibits the lowest Davies-Bouldin Index values among all methods, which is desirable. It reaches its minimum (around 0.55) for K=2, then slowly increases but still maintains a relatively low level compared to other methods. This suggests that despite its tendency to form chain-like clusters, this method can create clusters that are relatively compact and well-separated from each other, according to the Davies-Bouldin metric.
   - **AGNES (complete linkage)** shows a similar trend to AGNES (avg linkage), but with lower values. It reaches its minimum (around 1.0) for K=2, then increases to a maximum of about 1.2 for K=6, and then gradually decreases. Compared to single linkage, the values are higher, indicating less optimal clustering in terms of Davies-Bouldin.
   - **K-Means** behaves very similarly to AGNES (avg linkage) and Gaussian Mixture (diagonal/spherical). It achieves a high Davies-Bouldin Index value (around 1.4) for K=2, then increases to a maximum of about 1.85 for K=5, and then gradually decreases. High values indicate poor cluster quality in terms of Davies-Bouldin.
   - **Gaussian Mixture (diagonal)** shows a similar trend to K-Means and AGNES (avg linkage), reaching high Davies-Bouldin Index values. Initially, it's around 1.4 for K=2, then increases to a maximum of about 1.9 for K=5 or K=6, and then gradually decreases. High values indicate a lack of well-separated and compact clusters.
   - **Gaussian Mixture (spherical)** has an almost identical trajectory to Gaussian Mixture (diagonal) and K-Means, with very similar Davies-Bouldin Index values, reaching a maximum of about 1.9 for K=5 or K=6. This suggests that the assumptions of Gaussian mixture models (both diagonal and spherical) are not well-suited to the data structure in terms of the Davies-Bouldin Index.

3.0.2 Results for External Indices

Figure 2 illustrates the comparison of Adjusted Rand Index and Purity scores for K-Means, AGNES algorithms with different linkage methods (average linkage, complete linkage, single linkage) and Gaussian Mixture with different linkage methods (diagonal, spherical) as a function of the number of clusters k, ranging from 2 to 14.

1. **Adjusted Rand Index**

   - **K-Means**: Shows an increasing trend, achieving relatively high Adjusted Rand Index values, with several peaks (e.g., around 0.06 for $K = 8$ and $K = 12$). This suggests that K-Means is capable of identifying clustering structures that are moderately consistent

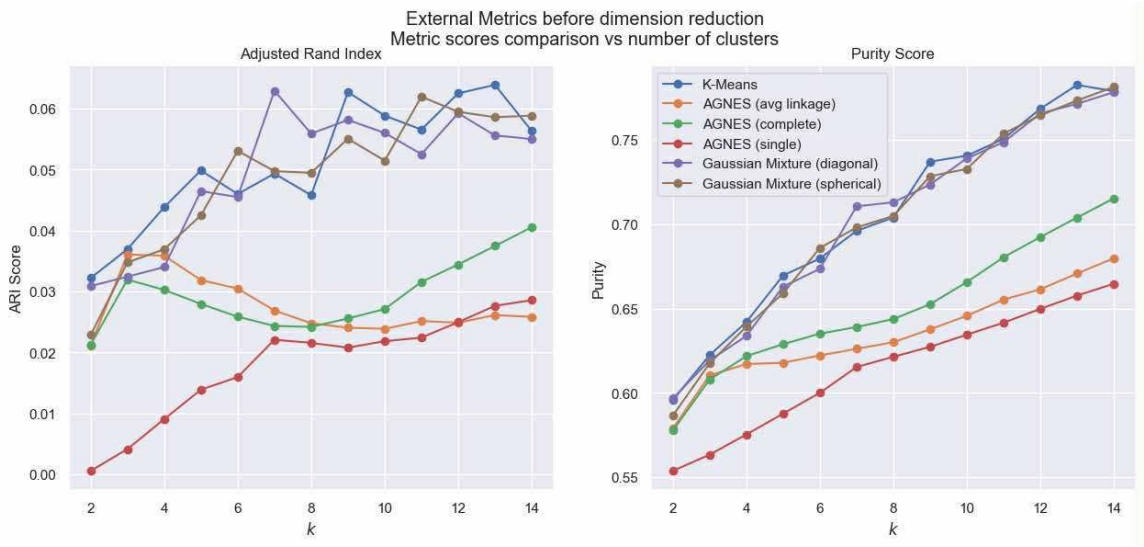

Figure 2: Index scores comparison for a given bootstrap iteration with corresponding best parameters

with the true labels, although the absolute values are still low, which might indicate difficulty in achieving a perfect match.

- **AGNES (avg linkage)**: Starts with lower values, exhibits some fluctuations, but generally maintains a moderate level. Its results are clearly lower than K-Means and Gaussian Mixture methods, oscillating around $0.03 - 0.04$. This suggests that this method has less agreement with the actual data divisions.
- **AGNES (complete linkage)**: Behaves similarly to AGNES (avg linkage), also achieving relatively low Adjusted Rand Index values (around $0.02 - 0.03$). This indicates a limited ability to group data in a way that is consistent with the reference labels.
- **AGNES (single linkage)**: This method consistently achieves the lowest Adjusted Rand Index values, often close to zero, especially for smaller K values. This indicates very poor agreement with the true labels, which is consistent with its tendency to form long, chain-like clusters that often do not correspond to natural groupings.
- **Gaussian Mixture (diagonal)**: Shows a similar pattern to K-Means, achieving high Adjusted Rand Index values, with peaks around 0.06. This suggests that Gaussian Mixture models with a diagonal covariance matrix are capable of capturing certain structures that align well with the true labels.
- **Gaussian Mixture (spherical)**: Has a very similar progression to Gaussian Mixture (diagonal) and K-Means, also achieving high Adjusted Rand Index values compared to AGNES methods. This indicates its ability to form clusters that are relatively well-aligned with the true data structure.

2. **Purity Score**

- **K-Means**: Shows a continuous increase in Purity Score as K increases, reaching some of the highest values (above 0.75 for larger K). This suggests that K-Means has a good ability to create clusters where points belonging to a single true class dominate, which is desirable in terms of purity.
- **AGNES (avg linkage)**: Shows a steady increase in Purity Score as K increases, although these values are lower than for K-Means and Gaussian Mixture methods, reaching about $0.65 - 0.70$ for larger K. This means that the clusters formed by this method are less "pure" in terms of the dominant class.
- **AGNES (complete linkage)**: Also shows an increase in Purity Score as K increases, but its values are lower than avg linkage, oscillating around $0.6 - 0.65$. This indicates lower cluster purity.
- **AGNES (single linkage)**: Achieves the lowest Purity Score values among all methods, although it shows a steady increase as K increases. This is consistent with earlier observations where this method performed poorly in aligning with the true labels. Its clusters are the most "mixed".
- **Gaussian Mixture (diagonal)**: Presents a very strong increase in Purity Score as K increases, reaching values comparable to, and even slightly higher than, K-Means for larger K (above 0.75). This suggests that Gaussian Mixture models with a diagonal covariance matrix are very effective in creating pure clusters.
- **Gaussian Mixture (spherical)**: Has an almost identical progression to Gaussian Mixture (diagonal) and K-Means, achieving very high Purity Score values (above 0.75) for larger K. This confirms that Gaussian Mixture models are generally good at forming clusters with high purity.

4 Dimensionality Reduction and Clustering

High-throughput biological data, such as gene expression matrices, typically consist of thousands of features (genes) and relatively few samples. This high dimensionality complicates both analysis and visualization. To uncover hidden structures and reduce computational complexity, we explore widely used dimensionality reduction technique: **Principal Component Analysis (PCA)**.

In this section, we evaluate how each technique contributes to understanding the structure of prostate tissue data and supports downstream clustering or classification efforts.

4.1 PCA tuning

Principal Component Analysis is a linear dimensionality reduction method that transforms the original dataset into a new coordinate system defined by orthogonal vectors (called principal components). These components are ordered by the amount of variance they explain in the data:

1. The first principal component captures the greatest variance.
2. Each subsequent component explains the next highest amount of variance under the constraint of being orthogonal to the previous ones.

PCA is commonly used to reduce noise, simplify data structures, and prepare datasets for further analysis such as clustering or classification. It assumes linear relationships and is most effective when the directions of greatest variance correspond to meaningful underlying structures.

To determine the optimal number of components, 50 bootstrap iterations were performed. Within each iteration, PCA was calculated for up to 30 components. The results of the PCA and component selection process are presented in Figure 3.

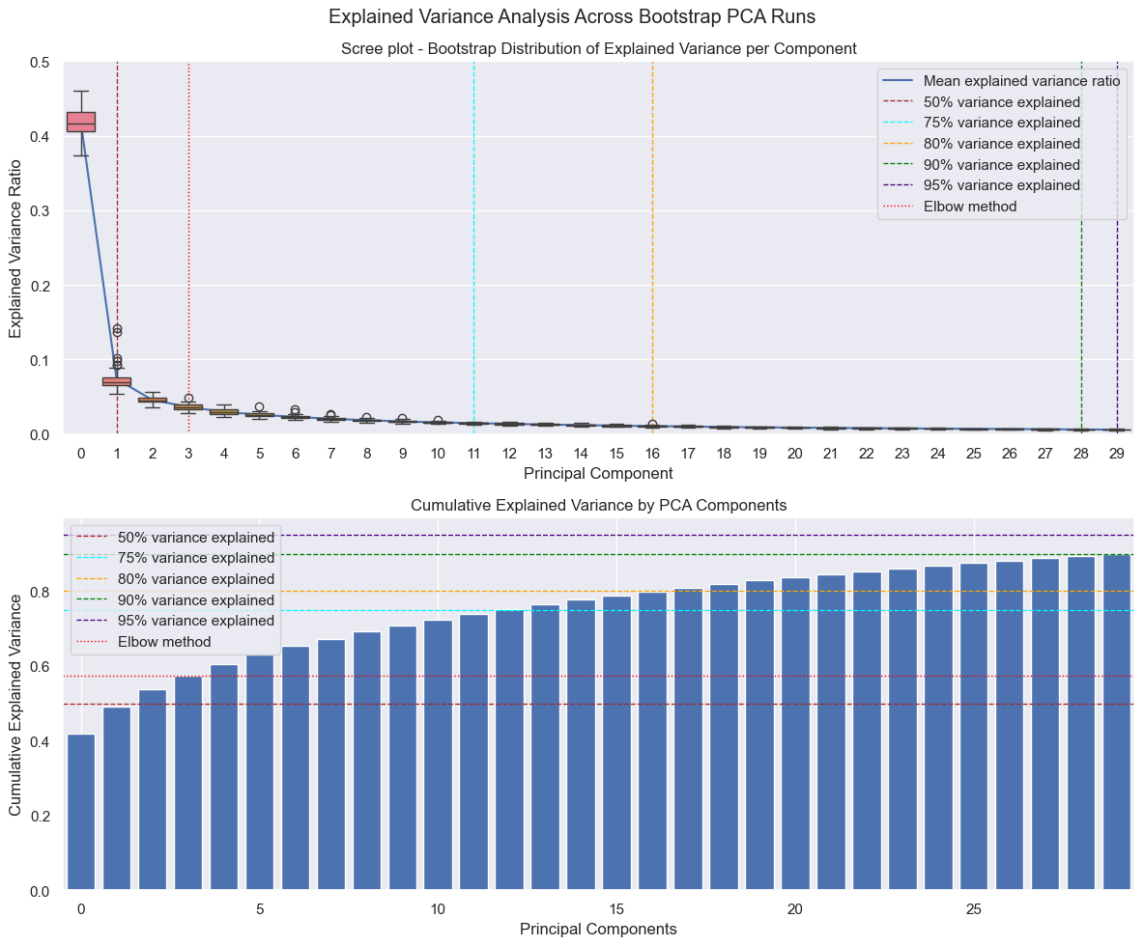

Figure 3: Explained variance analysis across bootstrap PCA runs.

In the upper plot, boxplots display the explained variance values obtained from the 50 bootstrap iterations. The highest interquartile range (IQR) can be observed for the first component, indicating greater variability in its explained variance across runs. The plot also shows the boundaries of explained variance percentages and highlights the threshold determined using the elbow method, which indicates the component after which additional components contribute minimally to the variance explained. The lower plot presents the cumulative explained variance alongside the aforementioned thresholds (for variance percentages and the elbow method). Several key conclusions

can be drawn:

- The first principal component (PC_1) alone explains approximately 40–45% of the total variance, while PC_2 and PC_3 contribute roughly 7% and 5% each.
- A clear "elbow" is visible at PC_3 or PC_4, indicating diminishing returns in explained variance beyond the third component.
- Achieving 50% cumulative explained variance requires only the first 2 PCs.
- Achieving 75% requires around 11 components.
- Achieving 80% requires approximately 16 components.
- Achieving 90% requires nearly all components (around 28–29).
- Achieving 95% effectively uses the full set of 29 PCs.

For further analysis, two scenarios were selected:

- A minimum of 50% variance explained, which corresponds to the first 2 principal components. This dimensionality is advantageous for visualization purposes.
- A minimum of 80% variance explained, requiring approximately 17 components, which provides better alignment with the underlying data structure and improved interpretability.

4.2 PCA with 2 components

The same bootstrap method as mentioned in Section 3 was performed. Results for Internal Indices are presented on Figure 4.

1. **Silhouette Index**:
   - Most methods exhibit similar behavior across different numbers of clusters. The highest Silhouette Index value is achieved by K-Means with 3 clusters. Overall, the best performance is observed for AGNES with the average linkage method, which consistently yields the highest scores for all $k > 3$.
   - The lowest scores across all cluster counts are obtained with AGNES using single linkage.
   - Compared to clustering without PCA, the results are more polarized, underscoring the ineffectiveness of AGNES with single linkage in this context.
2. **Davies-Bouldin Index**:
   - The results for K-Means and both Gaussian Mixture variants are very similar—a trend already observed and one that will appear again later. However, these methods exhibit the highest Davies-Bouldin Index values, which indicates poorer clustering since lower values are preferred for this metric.
   - AGNES with complete linkage also shows relatively high Davies-Bouldin values, suggesting suboptimal performance.

Figure 4: Internal Index plots for 2-component PCA.

- Interestingly, AGNES with average and single linkage methods perform best according to this metric, with single linkage achieving the lowest Davies-Bouldin scores. Unfortunately, other evaluation metrics do not reflect this trend, limiting the reliability of this finding.

External Index plots for 2-component PCA are shown on Figure 5.

1. **Adjusted Rand Index**:
   - The highest value is observed for AGNES with single linkage at $k = 4$.
   - Overall, all algorithms yield similar results across different cluster numbers.
   - The ARI values are generally low ($ARI \in (0.15, 0.40)$), indicating cluster assignments close to random labeling. This suggests that clustering may not be very effective for the analyzed dataset and, more broadly, for multiarray gene expression data.
2. **Purity Score**:
   - All methods show an increasing trend in purity scores as the number of clusters increases.
   - AGNES with single linkage consistently performs worst according to this metric.

Additionally, for 2 PCA components we can plot the values with reduced dimensions on a 2D plot, visible on Figure 6. Based on the results from Internal and External Indices values, the number of clusters were chosen accordingly:

- K-Means: 3 clusters,
- AGNES with single linkage: 2 clusters,

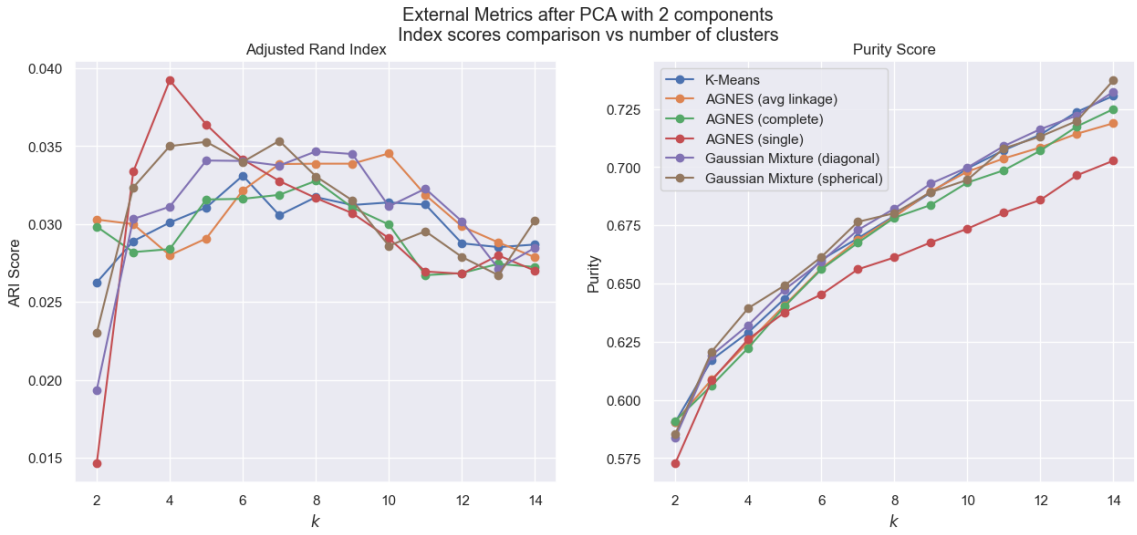

Figure 5: External Index plots for 2-component PCA.

- AGNES with average linkage: 3 clusters,
- AGNES with complete linkage: 3 clusters,
- Gaussian Mixture with diagonal covariance: 2 clusters,
- Gaussian Mixture with spherical covariance: 3 clusters.

The results illustrate the performance of each algorithm:

- **K-Means** performs reasonably well during clustering. Although cluster boundaries could be improved, the visibly distinct clusters were generally categorized correctly. Note the outlier in the upper-right corner of the plot; its classification varies by method. For K-Means, this outlier was assigned to cluster 0, together with the right-most points.
- **Gaussian Mixture (diagonal covariance)** behaves similarly to K-Means but better separates points within $PC_1 \in (-25, 0)$. With only two clusters, the outlier was grouped with its neighboring points.
- **AGNES (complete linkage)** yields clustering results comparable to those of K-Means and Gaussian Mixture (diagonal), representing one of the better clusterings observed.
- **AGNES (single linkage)** forms two clusters, placing nearly all points into one cluster and isolating the outlier in the second cluster. This represents the poorest clustering outcome.
- **AGNES (average linkage)** isolates the outlier in a separate cluster and divides the main points into two clusters. This single-point cluster may be useful for outlier detection, though overall clustering quality is moderate.

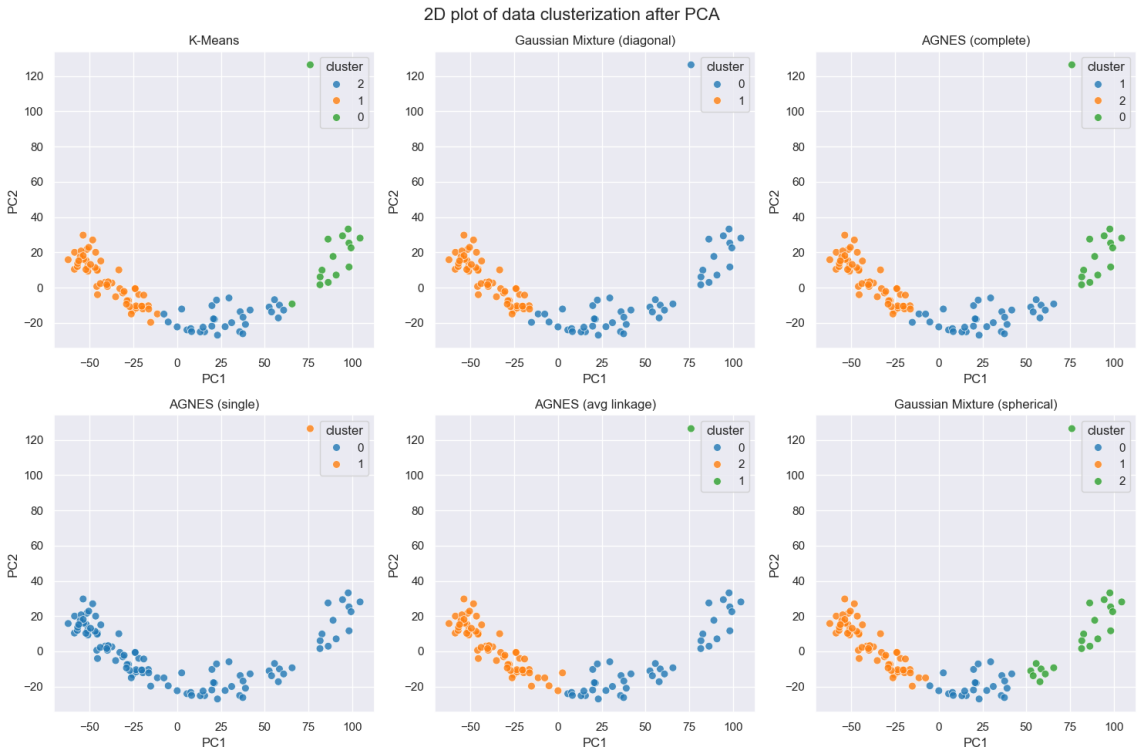

Figure 6: Data clusterization after 2-component PCA

- **Gaussian Mixture (spherical covariance)** offers a compromise between K-Means and AGNES with complete linkage, and can subjectively be considered among the best clusterings.

Using PCA with two components, data points can be mapped onto a 2D plane along with their corresponding class labels, allowing visual assessment of clustering performance. Figure 7 displays the class label assignment for each data point. The results are surprising and reveal no clear correlation between class labels and clusters. This outcome is expected since PCA and the clustering methods used are unsupervised and do not incorporate class information; thus, none of the methods effectively separate healthy and diseased samples.

This lack of correspondence may negatively impact subsequent classification tasks and is consistent with the low Adjusted Rand Index values observed. Additionally, the prostate dataset suffers from the curse of dimensionality, as the original 6033-dimensional data is unlikely to be faithfully represented in only two dimensions. Consequently, both the PCA projection and the resulting clusterings fail to reflect the true class structure, appearing more like random distributions.

Nevertheless, applying these methods provided valuable insights into the dataset's structure and highlighted the limitations of unsupervised dimensionality reduction and clustering in this context.

4.3 PCA with 17 components

The following sections present the clustering metric results obtained after applying dimensionality reduction using PCA to 17 components (explaining 80% of variance). They illustrate a comparison of algorithm behavior in the transformed feature space.

4.3.1 Internal Metrics (Silhouette Index and Davies-Bouldin Index)

Figure 8 displays the comparison of internal metrics.

- **Silhouette Index (left panel):** Analyzing the Silhouette Index values, we observe that AGNES (avg linkage) consistently maintains the highest values across a wide range of K, peaking around 0.50 for $K = 2$ and sustaining high values (e.g., 0.44 for $K = 12$). The other methods— K-Means, AGNES (complete linkage), and both Gaussian Mixture variants—show high initial values for $K = 2$ (around $0.40 - 0.49$), but drop sharply for larger K, reaching levels around $0.25 - 0.30$. AGNES (single linkage) is the weakest, with a rapid decline in values (to below 0.30) for most K.

- **Davies-Bouldin Index (right panel):** For the Davies-Bouldin Index (where lower values are better), AGNES (single linkage) surprisingly presents the lowest values across the entire range of K (around $0.40 - 0.45$), indicating optimally compact and well-separated clusters according to this metric. AGNES (avg linkage) also achieves relatively low values (around $0.60 - 0.85$). In contrast, K-Means, AGNES (complete linkage), and both Gaussian Mixture variants show significantly higher values (peaks above 1.3), which indicates poorer clustering quality in this regard.

4.3.2 External Metrics (Adjusted Rand Index and Purity Score)

Figure 9 illustrates the comparison of external metrics.

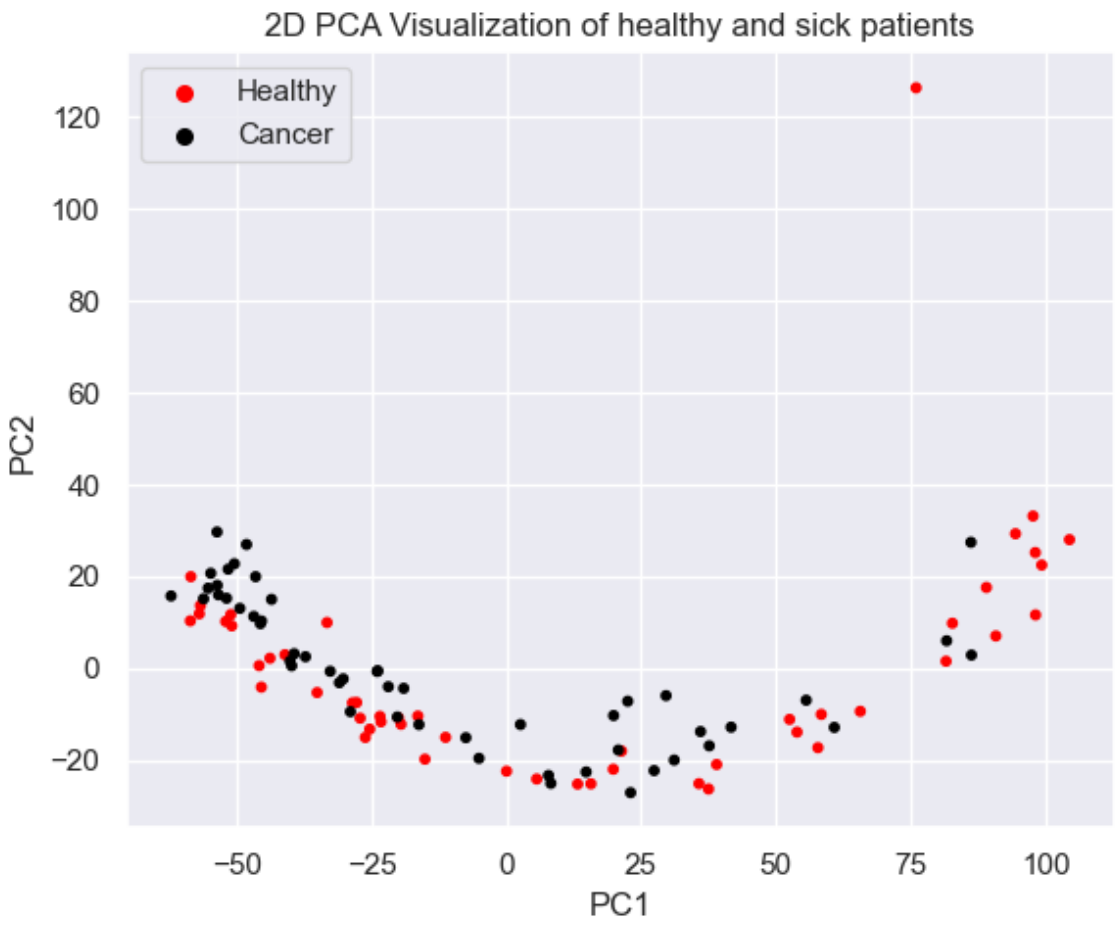

Figure 7: 2D PCA visualization of healthy and sick

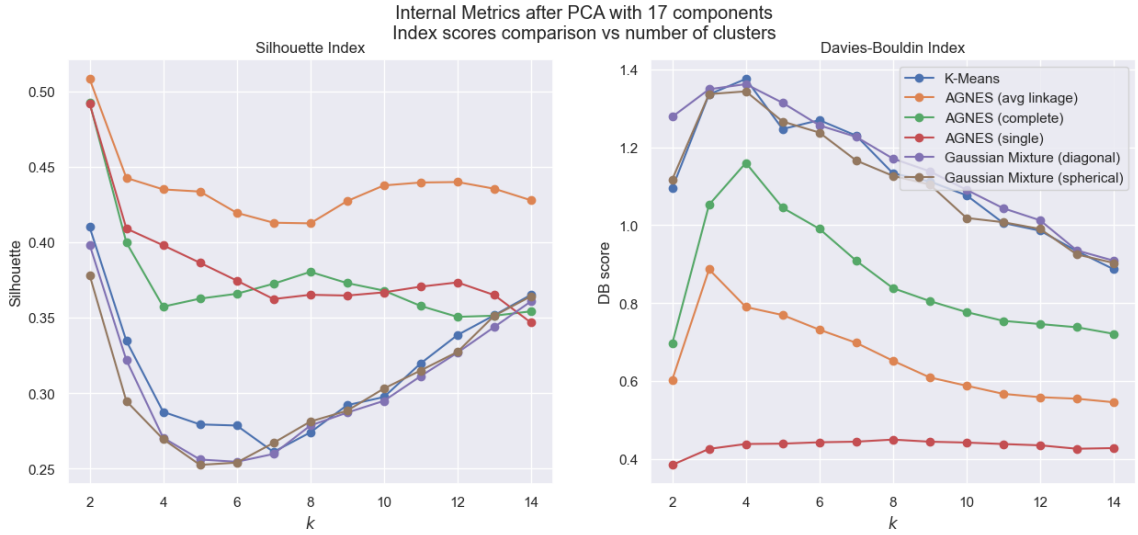

Figure 8: Comparison of internal clustering validation metrics (Silhouette Index and Davies-Bouldin Index) for data after dimensionality reduction using PCA to 17 components, as a function of the number of clusters K.

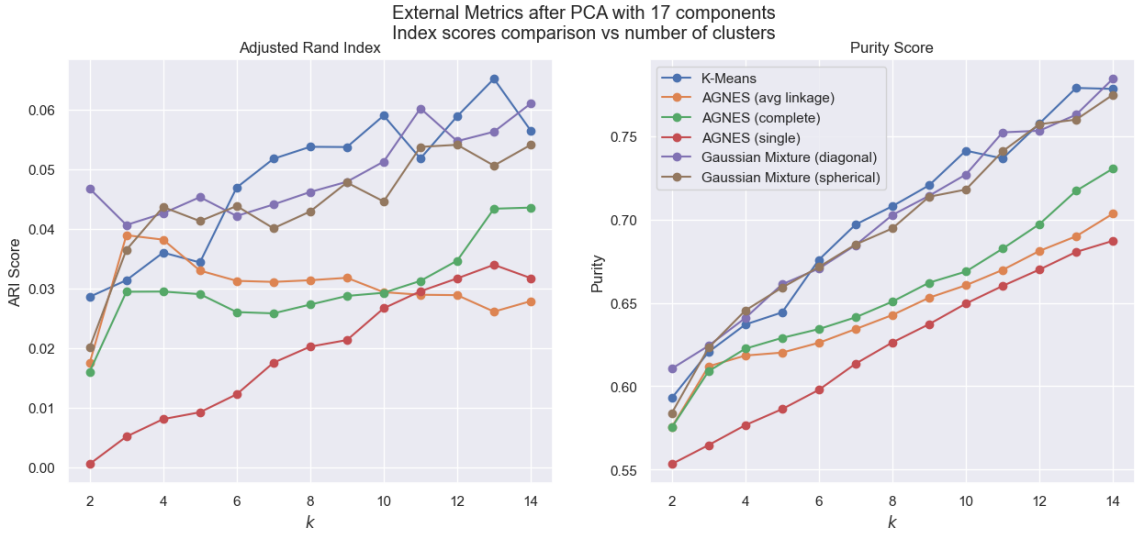

Figure 9: Comparison of external clustering validation metrics (Adjusted Rand Index and Purity Score) for data after dimensionality reduction using PCA to 17 components, as a function of the number of clusters K.

- **Adjusted Rand Index (left panel):** For the Adjusted Rand Index, K-Means and both Gaussian Mixture variants (diagonal and spherical) demonstrate the best and comparable results, oscillating between 0.04 and 0.065, with peak points for $K = 12$ and $K = 14$. This indicates their better agreement with true labels. AGNES (avg linkage) and AGNES (complete linkage) achieve moderate results $(0.03 - 0.04)$. AGNES (single linkage) is consistently the worst, with values close to zero for smaller K and a maximum of about 0.03 for $K = 14$. It should be emphasized that the overall Adjusted Rand Index values remain low, suggesting only a slight improvement over random class assignments.

- **Purity Score (right panel):** Regarding the Purity Score, K-Means and both Gaussian Mixture variants consistently achieve the highest values (above 0.75 for larger K, with maximum values around 0.78 for $K = 14$). This indicates their ability to form clusters in which points belonging to a single true class dominate. AGNES (avg linkage) and AGNES (complete linkage) present moderate results $(0.65 - 0.73)$, whereas AGNES (single linkage) is the weakest $(0.55 - 0.68)$.

5 Classification after PCA

5.1 LDA with PCA and GMM

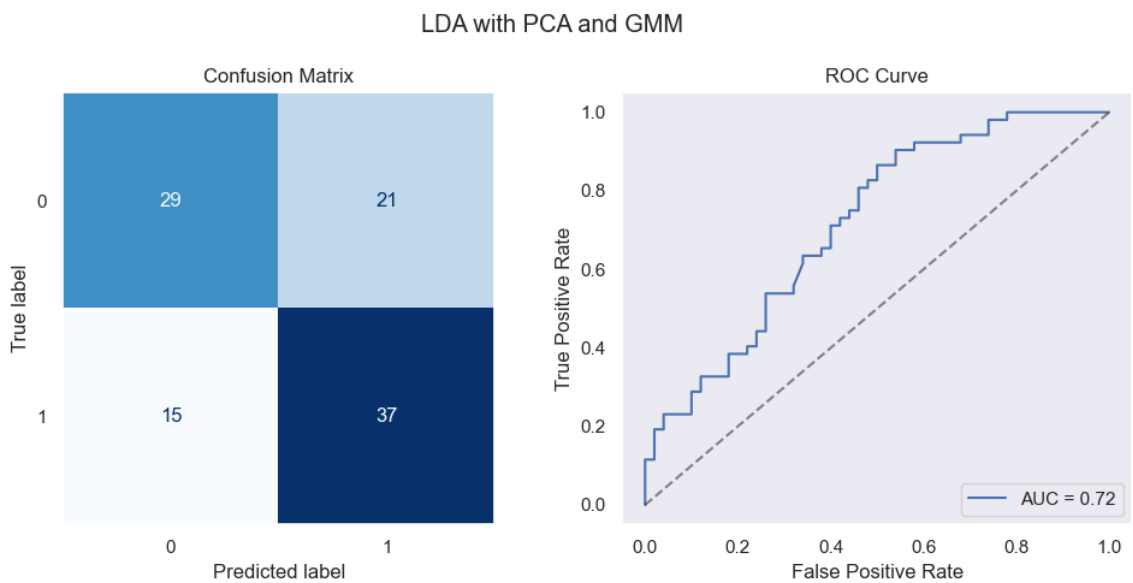

Figure 10: LDA after PCA and GMM.

Following the dimensionality reduction with Principal Component Analysis (PCA), a Gaussian Mixture Model (GMM) was employed for classification. The performance of this classifier was evaluated using a confusion matrix and a Receiver Operating Characteristic (ROC) curve analysis

to quantify its predictive accuracy and discriminative power. The results of this evaluation are presented in Figure 10.

The confusion matrix provides a granular view of the model's predictions against the true labels, breaking down the results into true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). The ROC curve illustrates the trade-off between the true positive rate (sensitivity) and the false positive rate across all classification thresholds, with the Area Under the Curve (AUC) serving as a single metric for overall separability. From this analysis, several key conclusions can be drawn:

- The model achieves a moderate overall accuracy of 64.7%. This indicates a classification performance that is better than random chance but leaves considerable room for improvement.
- The classifier's discriminative ability, as measured by the ROC analysis, results in an AUC score of 0.72. This value confirms a fair, but not strong, capacity to distinguish between the positive and negative classes.
- Analysis of the error types reveals a notable asymmetry. The model generated a high number of false positives (21), which consequently lowers the precision for the positive class to 63.8%. This suggests the model frequently misclassifies negative instances as positive.
- The model is comparatively more effective at identifying the actual positive cases, achieving a recall (sensitivity) of 71.2%. This indicates it successfully captures over two-thirds of all positive instances.
- The performance limitations are primarily attributed to a significant overlap between the class clusters within the reduced feature space created by PCA. At the boundary of these overlapping distributions, the GMM's probabilistic assignment is inherently ambiguous, leading to the observed classification errors.

5.2 LDA after PCA without GMM

For comparison, a second classification was performed using Linear Discriminant Analysis (LDA) directly on the features produced by the Principal Component Analysis (PCA), without the use of a Gaussian Mixture Model. The performance of this alternative pipeline was assessed using the same evaluation metrics. The resulting confusion matrix and ROC curve are shown in Figure 11.

This approach tests a different set of assumptions about the data structure, primarily that the classes are linearly separable within the PCA-reduced feature space. A detailed review of the classifier's performance and the underlying reasons for its effectiveness leads to several key conclusions:

- The model achieves a high overall accuracy of 86.3%. This marks a substantial improvement over the previous GMM-based model and indicates a strong alignment between the classification method and the data structure.
- The classifier's ability to distinguish between classes is excellent, as evidenced by an AUC score of 0.94. This value is close to a perfect score of 1.0 and signifies a very high degree of separability between the positive and negative classes.

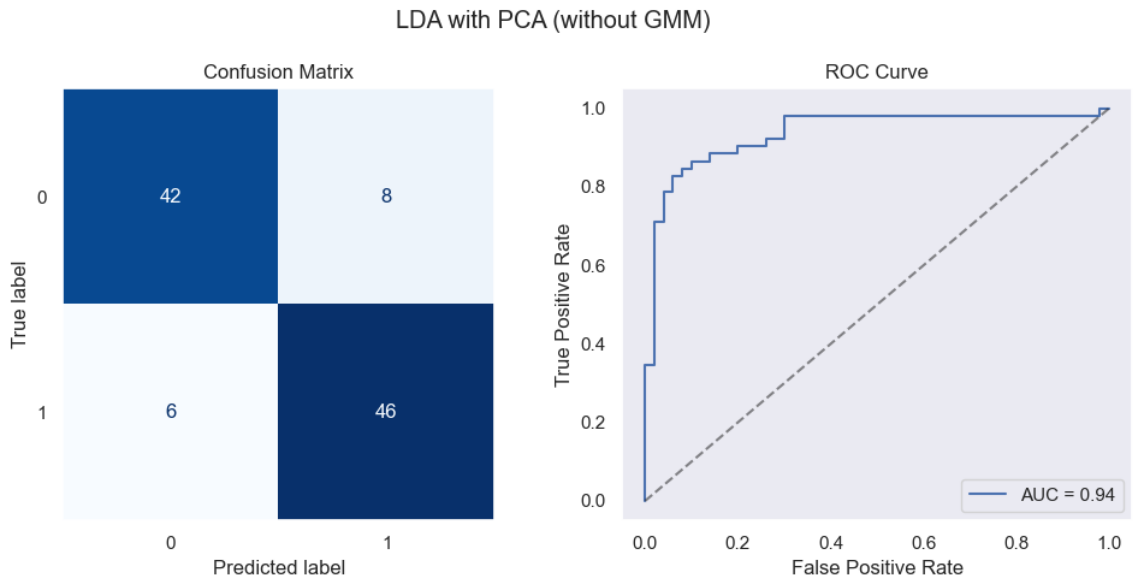

Figure 11: LDA after PCA.

- The error profile is significantly reduced and more balanced. The model produced only 8 false positives and 6 false negatives, demonstrating high performance in correctly classifying instances of both classes. This leads to a high precision of 85.2% and a high recall of 88.5%.
- The superior performance strongly suggests that the classes, within the feature space defined by PCA, are largely linearly separable. Unlike the GMM, which struggled with overlapping probabilistic clusters, the LDA classifier was able to find a highly effective linear decision boundary to separate the classes, resulting in fewer misclassifications.

5.3 Comparison to Part I

To evaluate the effectiveness of PCA as a pre-processing step, the performance of the PCA-LDA classifier was compared against a benchmark model from Part I of the project. This benchmark model utilized a feature selection algorithm to choose the most informative subset of original features before applying the same Linear Discriminant Analysis classifier. The results for this feature selection-based model are presented in Figure 12.

The comparison aims to determine which dimensionality reduction strategy—feature transformation (PCA) or feature selection—provided a more optimal feature space for classification by LDA.

- The LDA model with feature selection is the superior performer across all key metrics. It achieved a higher overall accuracy of 91.2% compared to the 86.3% from the PCA-LDA model, and its AUC score of 0.95 is marginally higher than the PCA-LDA's 0.94.
- The error rate for the feature selection model is substantially lower. It made only 9 misclassifications in total (4 false positives, 5 false negatives), whereas the PCA-LDA model made

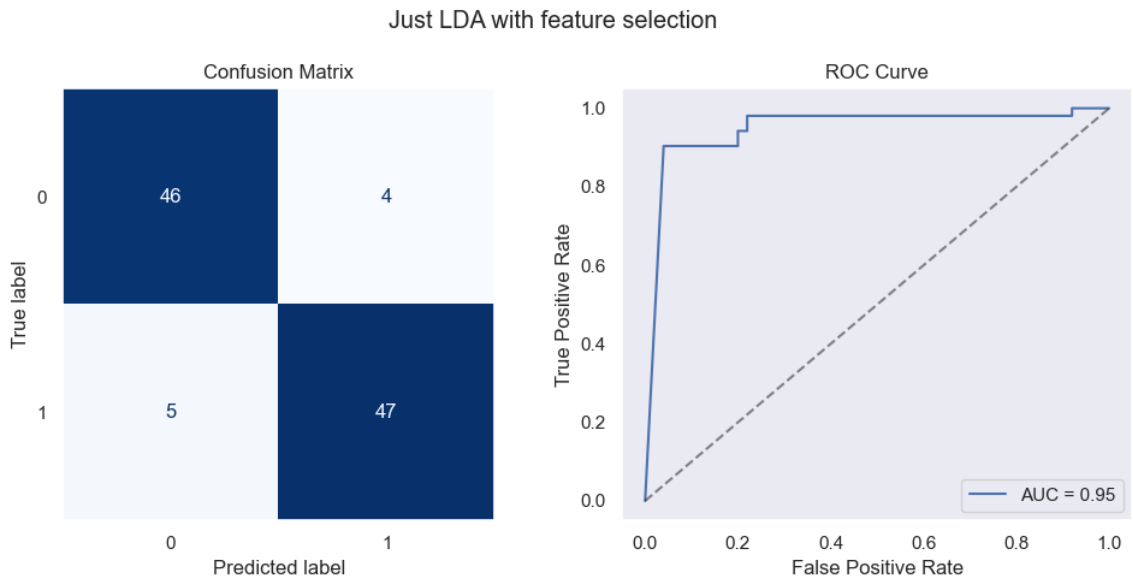

Figure 12: LDA after feature selection.

14 misclassifications (8 false positives, 6 false negatives). This indicates a more robust and reliable classification boundary.

- The underlying reason for this performance difference lies in the nature of the methods. Feature selection identifies and retains the most powerful original features, preserving their inherent discriminatory information. In this case, it appears that a specific subset of the original variables was highly effective for linear separation.
- In contrast, PCA transforms the entire feature set into new, composite components. While this method successfully captured the majority of the variance needed for classification, the transformation process likely resulted in a slight loss of separability compared to using the best original features directly. For this dataset, selecting the right features was a more effective strategy than creating new ones.

6 Conclusions

The analysis of the prostate cancer gene expression dataset, which contains 6033 gene features for 102 samples, demonstrated the complexities of applying unsupervised learning methods to high-dimensional biological data. The investigation utilized a variety of clustering algorithms (K-Means, AGNES, Gaussian Mixture Models) and Principal Component Analysis (PCA) for dimensionality reduction.

The core findings of the report indicate that unsupervised methods did not yield optimal results in clearly separating the data according to the true class labels (cancerous vs. non-cancerous). The clustering results, when evaluated with external indices like the Adjusted Rand Index, showed values

close to what would be expected from random chance, suggesting a weak correlation between the generated clusters and the actual biological classes. This was further confirmed when, after reducing the data to two principal components, a 2D visualization showed no clear separation between healthy and diseased samples.

Despite these limitations, the process provided invaluable insights. The application of various clustering algorithms and evaluation metrics highlighted that different methods capture unique aspects of the data's structure and that no single algorithm is universally superior. For instance, while AGNES with single linkage performed poorly on most metrics, it consistently showed the best (lowest) scores on the Davies-Bouldin Index, a contradiction that underscores the importance of using multiple validation techniques.

Furthermore, the exploration of PCA as a pre-processing step for classification revealed its utility and its limitations. While an LDA classifier applied after PCA achieved a respectable accuracy of 86.3% and an AUC of 0.94, it was outperformed by the LDA model from Part I of the project, which used feature selection instead of PCA. The feature selection approach yielded a higher accuracy of 91.2% and an AUC of 0.95, suggesting that for this particular dataset, retaining the most informative original genes was a more effective strategy than transforming them into composite components.

In essence, while the unsupervised methods did not successfully uncover the underlying class structure on their own, their application was a critical step in understanding the dataset's characteristics, the curse of dimensionality, and the behaviors of the algorithms themselves.